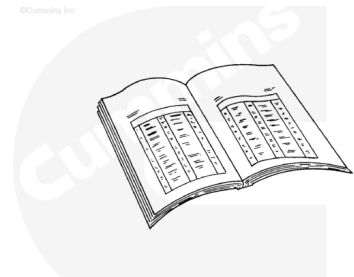


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

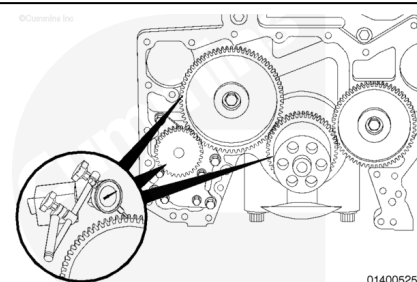
- Disconnect the batteries and air starter to prevent accidental starting.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>).
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Remove the fan belt. Refer to Procedure 008-002 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the alternator belt and alternator. Refer to Procedure 013-001 in Section 13 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Remove the fan belt tensioner. Refer to Procedure 008-036 in Section 8 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>).
- Remove the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>) in Section 1.

- Remove the front crankshaft seal. Refer to Procedure 001-023 (/qs3/pubsys2/xml/en/procedures/89/89-001-023.html) in Section 1.
- Remove the alternator drive gear and shaft. Refer to Procedure 009-009 (/qs3/pubsys2/xml/en/procedures/89/89-009-009.html) in Section 9.
- Remove the front engine mount. Refer to Procedure 016-010 (/qs3/pubsys2/xml/en/procedures/00/00-016-010.html) in Section 16.
- Remove the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html) in Section 1.

Measure

⚠ CAUTION ⚠

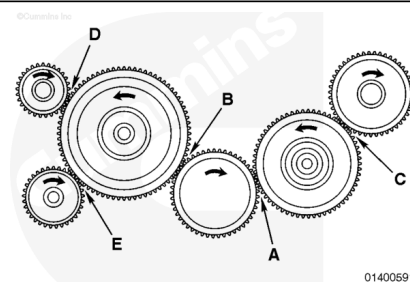
The adjacent (engaging) gear must not turn as the gear being measured is turned.



Position a dial indicator so the tip is contacting the surface of the gear tooth as shown. Do **not** allow the mating gear to turn.

Rotate the gear being measured **clockwise**. Position the indicator to zero, rotate the gear **counterclockwise**, and read the indicator.

Measure the gear backlash.

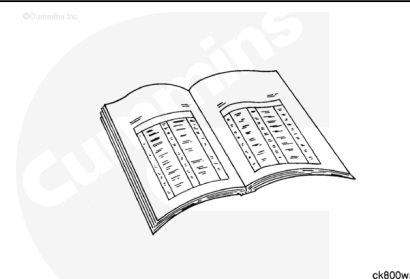


Gear Train Backlash		
	MIN	MAX
A	0.144 mm [0.006 in]	0.320 mm [0.013 in]
B	0.134 mm [0.005 in]	0.362 mm [0.014 in]
C	0.114 mm [0.004 in]	0.320 mm [0.013 in]
D	0.121 mm [0.005 in]	0.333 mm [0.013 in]

Gear Train Backlash		
	MIN	MAX
E	0.121 mm [0.005 in]	0.333 mm [0.013 in]

Finishing Steps

- Install the front cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Install the front engine mount. Refer to Procedure 016-010 (</qs3/pubsys2/xml/en/procedures/00/00-016-010.html>).
- Install the alternator drive gear and shaft. Refer to Refer to Procedure 009-009 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>).
- Install the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>).
- Install the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Install the fan belt tensioner. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>).
- Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Install the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>).
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Install new oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/89/89-007-013.html>).
- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Connect the batteries.
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>).
- Operate the engine to normal operating temperature. Check for leaks.



Last Modified: 06-May-2015
